

underroot of a number

$$\sqrt{25} = 5$$

$$\sqrt{16} = 4$$

$$\sqrt{(1-\sqrt{2})^2} = 1 - \sqrt{2}$$

x

$$\begin{aligned}\sqrt{(1-\sqrt{2})^2} &= \sqrt{(\sqrt{2}-1)^2} \\ &= \sqrt{2}-1 \checkmark\end{aligned}$$

$$\begin{array}{l|l}\sqrt{x} = 3 & \underline{Q} \quad \sqrt{x} = -7 \\ x = 9 & \text{no solution}\end{array}$$

$$\sqrt{x} + \sqrt{y} = -5 \quad x$$

$$\sqrt{x} - \sqrt{y} = -4 \quad \checkmark$$

* underroot of a non-negative number is always non negative

ie $\sqrt{x} \geq 0$

if $x^2 = 9$ then $x = 3$ (F)	if $x^2 = 9$ then $x = \pm 3$ (T)	if $x = \sqrt{9}$ then $x = 3$ (T)	if $x = \sqrt{9}$ then $x = \pm 3$ (F)
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$$x^2 = 9$$

$$\begin{aligned}
 x^2 &= 9 \\
 x^2 - 3^2 &= 0 \\
 (x-3)(x+3) &= 0 \\
 x-3=0 & \mid x+3=0 \\
 x=3 & \mid x=-3
 \end{aligned}$$

$$\begin{aligned}
 &\rightarrow x^2 = 16 \\
 &x = \sqrt{16} \\
 &= \pm 4
 \end{aligned}$$

$$\begin{aligned}
 &\rightarrow x^2 = 16 \\
 &x = \sqrt{16} \\
 &= 4
 \end{aligned}$$

$$\begin{aligned}
 &\rightarrow x^2 = 16 \\
 &x = \pm \sqrt{16} \\
 &= \pm 4
 \end{aligned}$$

correct
 (method - incorrect)

Incorrect

ib. $\sqrt{x+2} = x$, find x

Soln $x+2 = x^2$ (sbs)

$$x^2 - x - 2 = 0$$

$$x^2 - 2x + x - 2 = 0$$

$$x(x-2) + 1(x-2) = 0$$

$$(x+1)(x-2) = 0$$

$$x = -1 \quad \times \quad | \quad x = 2 \quad \checkmark$$

$$\sqrt{-1+2} = -1$$

$$\sqrt{1} = -1$$

$\times$

$$\sqrt{2+2} = 2$$

$$\sqrt{4} = 2 \quad \checkmark$$

$$\frac{1}{0} = \text{undefined}$$

$$\frac{1}{0} = a$$

$$1 = 0 \times a$$

$$a = \text{not possible}$$

$$\frac{0}{0} = \text{undetermined}$$

$$\frac{0}{0} = b$$

$$0 = \underline{b \times 0}$$

$$b = \text{anything}$$

Q 9b $x + \frac{1}{x} = 14$, then find

$$\sqrt{x + \frac{1}{x}} = ?$$

$$\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 = x + \frac{1}{x} + 2 \cdot \sqrt{x} \cdot \frac{1}{\sqrt{x}}$$

$$= x + \frac{1}{x} + 2$$

$$= 14 + 2$$

$$\Rightarrow \left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 = 16$$

$$\Rightarrow \sqrt{x} + \frac{1}{\sqrt{x}} = \pm \sqrt{16}$$

$$= \pm 4$$

$$\sqrt{x} + \frac{1}{\sqrt{x}} = 4$$

✓

$$\sqrt{x} + \frac{1}{\sqrt{x}} = -4$$

not possible as sum of two non negative numbers can not be negative

Completing square method $(ax^2 + bx + c)$

① take a common

② write in form $(x + \alpha)^2 + \beta$

$$\left(x + \frac{\text{coeff of } x}{2}\right)^2 + \left[\text{const term} - \left(\frac{\text{coeff of } x}{2}\right)^2\right]$$

Q Apply completing the square on

a) $x^2 - 8x + 12$

b) $3x^2 + 4x - 5$

$$2x^2 - 3x - 4$$

$$2\left(x^2 - \frac{3x}{2} - 2\right)$$

$$2\left[\left(x - \frac{3}{4}\right)^2 + \left(-2 - \left(\frac{3}{4}\right)^2\right)\right]$$

$$2\left[\left(x - \frac{3}{4}\right)^2 + \left(-2 - \frac{9}{16}\right)\right]$$

$$2\left[\left(x - \frac{3}{4}\right)^2 - \frac{41}{16}\right]$$

$$2\left(x - \frac{3}{4}\right)^2 - \frac{41}{8}$$

$$3x^2 + 4x - 5$$

$$3 \left[x^2 + \frac{4}{3}x - \frac{5}{3} \right]$$

$$3 \left[\left(x + \frac{2}{3} \right)^2 + \left(-\frac{5}{3} - \left(\frac{2}{3} \right)^2 \right) \right]$$

$$3 \left[\left(x + \frac{2}{3} \right)^2 + \left(-\frac{5}{3} - \frac{4}{9} \right) \right]$$

$$3 \left[\left(x + \frac{2}{3} \right)^2 + \frac{-15 - 4}{9} \right]$$

$$3 \left[\left(x + \frac{2}{3} \right)^2 - \frac{19}{9} \right]$$

$$x^2 + y^2 = 0$$

$$x=0 \text{ and } y=0$$

$$x^2 + y^2 + z^2 = 0$$

$$x=0, y=0, z=0$$

eg

$$(x-1)^2 + (y+2)^2 = 0$$

Soln

$$\begin{array}{c|c} x-1=0 & y+2=0 \\ \hline x=1 & y=-2 \end{array}$$

Q $x^2 + y^2 + 4x - 6y + 13 = 0$, then find x and y .

$$x^2 + y^2 + 4x - 6y + 13 = 0$$

$$\underline{(x^2 + 4x)} + \underline{(y^2 - 6y)} + 13 = 0$$

$$= (x+2)^2 - 4 + (y-3)^2 - 9 + 13 = 0$$

$$\Rightarrow (x+2)^2 + (y-3)^2 = 0$$

$$x+2=0 \text{ and } y-3=0$$

$$\therefore x = -2 \text{ and } y = 3$$

$$\begin{array}{l} x^2 + 4x \\ \left(x + \frac{4}{2}\right)^2 + (0 - 2^2) \end{array}$$

$$(x+2)^2 - 4$$

$$\begin{array}{l} y^2 - 6y \\ \left(y + \frac{-6}{2}\right)^2 + (0 - (-3)^2) \end{array}$$

$$(y-3)^2 - 9$$

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)(a^2 + b^2 + c^2 - ab - bc - ca)$$

$$a^3 + b^3 + c^3 - 3abc = \frac{1}{2}(a+b+c)[(a-b)^2 + (b-c)^2 + (c-a)^2]$$

if $a+b+c=0$, then $a^3 + b^3 + c^3 = 3abc$

if $a^3 + b^3 + c^3 - 3abc = 0$, then $a+b+c=0$ or $a=b=c$

Q 9 If $x+2y+3z=0$ and $x^3+8y^3+27z^3 = kxyz$, then find $k=?$

Q simplify $\frac{(a-b)^3 + (b-c)^3 + (c-a)^3}{(a-b)(b-c)(c-a)} = ?$

Soln

$$x + 2y + 3z = 0$$

then $(x)^3 + (2y)^3 + (3z)^3 = 3(x)(2y)(3z)$

$$x^3 + 8y^3 + 27z^3 = 18xyz$$

$$k = 18$$

Soln

$$\frac{(a-b)^3 + (b-c)^3 + (c-a)^3}{(a-b)(b-c)(c-a)}$$

$$a-b = x$$

$$b-c = y$$

$$c-a = z$$

$$\cancel{a-b} + \cancel{b-c} + \cancel{c-a} = x + y + z$$

$$x + y + z = 0$$

$$= \frac{x^3 + y^3 + z^3}{xyz}$$

$$= \frac{3\cancel{x}\cancel{y}\cancel{z}}{\cancel{xyz}}$$

$$= 3$$

factorise

$$x^4 + x^2 + 1$$

$$= \frac{x^4 + 2x^2 + 1}{1} - x^2$$

$$= (x^2 + 1)^2 - (x)^2$$

$$= (x^2 + 1 + x)(x^2 + 1 - x)$$

$$= (x^2 + x + 1)(x^2 - x + 1)$$

$$x^4 + 6x^2 + 25$$

$$\underbrace{x^4 + 10x^2 + 25}_{0} - 4x^2$$

$$(x^2 + 5)^2 - (2x)^2$$

$$(x^2 + 5 + 2x)(x^2 + 5 - 2x)$$

$$(x^2 + 2x + 5)(x^2 - 2x + 5)$$

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Exponents and surds

$$a) a^m \cdot a^n = a^{m+n}$$

$$b) a^m \div a^n = a^{m-n}$$

$$c) (a^m)^n = a^{mn}$$

$$d) a^m \cdot b^m = (ab)^m$$

$$e) \left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$$

$$f) a^0 = 1$$

$$g) a^{-1} = \frac{1}{a}, \quad \left(\frac{a}{b}\right)^{-1} = \frac{b}{a}, \quad a^{-m} = \frac{1}{a^m}$$

$$h) \sqrt[n]{a} = a^{\frac{1}{n}}, \quad \sqrt{a} = a^{\frac{1}{2}}$$

$$i) \sqrt[n]{a^m} = (a^m)^{\frac{1}{n}} = a^{\frac{m}{n}}$$

$$(64)^{\frac{3}{2}}$$

$$64 = 2^6$$

$$= (2^6)^{\frac{3}{2}}$$

$$= 2^{6 \times \frac{3}{2}}$$

$$= 2^9$$

Simplify

$$(i) 9^{-\frac{1}{2}} + 625^{\frac{1}{4}} + 8^{\frac{5}{3}} \quad | \quad (ii) \sqrt[3]{\sqrt[3]{\sqrt[4]{64}}}$$

$$\begin{aligned}
 & 9^{-1/2} + 625^{1/4} + 8^{5/3} \\
 & (3^2)^{-1/2} + (5^4)^{1/4} + (2^3)^{5/3} \\
 & = 3^{-1} + 5^1 + 2^5 \\
 & = \frac{1}{3} + 5 + 32 \\
 & = \frac{1}{3} + 37 \\
 & = \frac{112}{3}
 \end{aligned}$$

$$\begin{aligned}
 & \sqrt[3]{\sqrt[4]{64}} \\
 & = \left(\left((64)^{1/4} \right)^{1/3} \right)^{1/2} \\
 & = \left(\left((2^6)^{1/4} \right)^{1/3} \right)^{1/2} \\
 & = 2^{\cancel{6} \times \frac{1}{4} \times \frac{1}{3} \times \frac{1}{2}} \\
 & = 2^{1/4}
 \end{aligned}$$

simplify

$$a) a \left(\frac{\sqrt{a} + \sqrt{b}}{2b\sqrt{a}} \right)^{-1} + b \left(\frac{\sqrt{a} + \sqrt{b}}{2a\sqrt{b}} \right)^{-1}$$

$$b) \frac{2 \cdot 3^{n+1} - 7 \cdot 3^{n-1}}{3^{n+1} + 2 \left(\frac{1}{3} \right)^{1-n}}$$

soln a) $a \frac{2b\sqrt{a}}{\sqrt{a} + \sqrt{b}} + b \frac{2a\sqrt{b}}{\sqrt{a} + \sqrt{b}}$

$$\frac{2ab\sqrt{a}}{\sqrt{a} + \sqrt{b}} + \frac{2ab\sqrt{b}}{\sqrt{a} + \sqrt{b}}$$

$$\frac{2ab\sqrt{a} + 2ab\sqrt{b}}{\sqrt{a} + \sqrt{b}}$$

$$= \frac{2ab(\cancel{\sqrt{a} + \sqrt{b}})}{\cancel{\sqrt{a} + \sqrt{b}}} = 2ab$$

$$\frac{2 \cdot 3^{n+1} - 7 \cdot 3^{n-1}}{3^{n+1} + 2 \left(\frac{1}{3}\right)^{1-n}}$$

$$= \frac{2 \cdot 3^n \cdot 3^1 - 7 \cdot \frac{3^n}{3^1}}{3^n \cdot 3^1 + 2 \cdot \frac{3^n}{3^1}}$$

$$= \frac{\cancel{3^n} \left(6 - \frac{7}{3}\right)}{\cancel{3^n} \left(3 + \frac{2}{3}\right)}$$

$$= \frac{\cancel{\frac{11}{3}}}{\cancel{\frac{11}{3}}}$$

$$= 1$$

$$\left(\frac{1}{3}\right)^{1-n}$$

$$\left(3^{-1}\right)^{1-n}$$

$$3^{(-1)(1-n)}$$

$$3^{-1+n}$$

$$3^{n-1}$$

$$\frac{3^n}{3^1}$$

Q write

$$2^{\frac{1}{3}} 3^{\frac{1}{2}} = (\alpha)^{\frac{1}{n}} \text{ form}$$

and also find α .

$$2^{\frac{1}{3}} 3^{\frac{1}{2}}$$

$$2^{\frac{2}{6}} 3^{\frac{3}{6}}$$

$$\left(2^2\right)^{\frac{1}{6}} \left(3^3\right)^{\frac{1}{6}}$$

$$\left(2^2 \times 3^3\right)^{\frac{1}{6}}$$

$$= (108)^{\frac{1}{6}}$$

$$\alpha = 108$$

square roots

Q find square root of $4+2\sqrt{3}$

Soln
let, $\sqrt{4+2\sqrt{3}} = \sqrt{a} + \sqrt{b}$

$$4+2\sqrt{3} = (\sqrt{a} + \sqrt{b})^2$$

$$4+2\sqrt{3} = a+b+2\sqrt{ab}$$

compare	$4 = a+b$	$2\sqrt{3} = 2\sqrt{ab}$
		$3 = ab$

$$a=1, b=3$$

$$\therefore \sqrt{4+2\sqrt{3}} = \sqrt{1} + \sqrt{3} = 1 + \sqrt{3}$$

Q find square root of $5 + \sqrt{24}$

Q Rationalise the denominator

$$\frac{1}{1 + \sqrt{2} + \sqrt{3}}$$

$$\sqrt{5 + \sqrt{24}} = \sqrt{a} + \sqrt{b}$$

$$5 + \sqrt{24} = a + b + 2\sqrt{ab}$$

$$a + b = 5 \quad | \quad \sqrt{24} = 2\sqrt{ab}$$

$$\cancel{4}\sqrt{6} = \cancel{2}\sqrt{ab}$$

$$6 = ab$$

$$a = 2, b = 3$$

$$\sqrt{5 + \sqrt{24}} = \sqrt{2} + \sqrt{3} //$$

$$\frac{1}{1 + \sqrt{2} + \sqrt{3}}$$

$$\frac{(1 + \sqrt{2} - \sqrt{3})}{(\underline{1 + \sqrt{2} + \sqrt{3}})(\underline{1 + \sqrt{2} - \sqrt{3}})}$$

$$= \frac{1 + \sqrt{2} - \sqrt{3}}{(1 + \sqrt{2})^2 - \sqrt{3}^2}$$

$$= \frac{1 + \sqrt{2} - \sqrt{3}}{\cancel{1} + \cancel{2} + 2\sqrt{2} - \cancel{3}}$$

$$= \frac{1 + \sqrt{2} - \sqrt{3}}{2\sqrt{2}}$$

$$= \frac{(1 + \sqrt{2} - \sqrt{3})\sqrt{2}}{2\sqrt{2}\sqrt{2}}$$

$$= \frac{\sqrt{2} + 2 - \sqrt{6}}{4}$$

Q 96

a and b are rational number

$$(a + b\sqrt{2})(3 + 2\sqrt{2}) = 4 + 5\sqrt{2}, \text{ then find } a \text{ and } b.$$

$$a + b\sqrt{2} = \frac{4 + 5\sqrt{2}}{3 + 2\sqrt{2}}$$

$$= \frac{(4 + 5\sqrt{2})(3 - 2\sqrt{2})}{(3 + 2\sqrt{2})(3 - 2\sqrt{2})}$$

$$= \frac{12 - 8\sqrt{2} + 15\sqrt{2} - 20}{3^2 - (2\sqrt{2})^2}$$

$$= \frac{-8 + 7\sqrt{2}}{1}$$

$$a + b\sqrt{2} = -8 + 7\sqrt{2}$$

$$a = -8$$

$$b = 7$$

Surds: any number in the form of $a^{\frac{1}{n}}$, where a is rational and n is natural is called surds (provided overall quantity is irrational)

$$3^{\frac{1}{2}} \left\{ \begin{array}{l} 8^{\frac{2}{3}} \\ = 4 \times \end{array} \right. \left| \begin{array}{l} \text{rational} \pm \text{surds} = \text{surds} \\ \text{surds} \pm \text{surds} = \text{surds} \end{array} \right| \begin{array}{l} 1 + \sqrt{2} \\ \sqrt{2} - \sqrt{3} \end{array}$$

Q write the following surds in increasing order

$$2^{\frac{1}{2}}, 3^{\frac{1}{3}}, 5^{\frac{1}{4}}$$

$$\text{LCM}(2, 3, 4) = 12$$

Soln

$$\left(2^{\frac{1}{2}}, 3^{\frac{1}{3}}, 5^{\frac{1}{4}} \right) \left| \begin{array}{l} \left(2^{\frac{1}{2}} \right)^{12} \\ = 2^6 \\ = 64 \end{array} \right| \begin{array}{l} \left(3^{\frac{1}{3}} \right)^{12} \\ = 3^4 = 81 \end{array} \left| \begin{array}{l} \left(5^{\frac{1}{4}} \right)^{12} \\ = 5^3 = 125 \end{array} \right.$$

$$64 < 81 < 125$$

$$\boxed{2^{\frac{1}{2}} < 3^{\frac{1}{3}} < 5^{\frac{1}{4}}}$$

$$\begin{array}{ccc}
 2^{2/3} & 3^{1/2} & 5^{1/3} \\
 \left(2^{2/3}\right)^6 & \left(3^{1/2}\right)^6 & \left(5^{1/3}\right)^6 \\
 = 2^4 & = 3^3 & = 5^2 \\
 16 & = 27 & = 25
 \end{array}$$

$$16 < 25 < 27$$

$$2^{2/3} < 5^{1/3} < 3^{1/2}$$

$$\text{LCM}(3, 2, 3) = 6$$

$$= \frac{1}{0.0001} =$$